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Ultra-Low Phase Noise, Digital Transceiver for High Range Resolution Pulsed Agile Radar

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***Abstract*— In the modern age, highly sensitive advanced radar systems require very stable and accurate radar waveforms for micro-Doppler signature analysis and smaller Radar Cross section (RCS) detection. Also, to combat RADAR jammers and in naval applications, wideband radars operate at multiple spot frequencies in pulse agility mode.**

This paper discusses designing a wide-band digital transceiver using RF DAC and ADC with ultra-low phase noise and High-speed switching for pulsed agile radar. Two methods using RF DAC are being discussed and compared for low-phase noise waveform generation. Switching times of less than 500ns have been achieved for the transmit pulse in the agility mode.

I. INTRODUCTION

In the last few years, technological advancement in Radar systems demanded ultra-low phase noise transmission, programmable digital transmit and receive module (DTRM) to directly convert the RF to digital at the antenna array level and send the digital baseband signal through optical/ Ethernet for further signal processing.

Phase noise is one of the most important parameters for the design of DTRM. Since the radar determines time delay, and therefrom range, by counting clock cycles, as determined by their edges, any randomness in the clock timing translates to randomness in the intended timing measurement.

The two sources of phase noise in the radar system are

A. The phase noise in the oscillator

B. Additive phase noise from frequency translation circuits such as Phase-Locked Loops (PLLs) and RF DAC in modern radars.

The techniques of PLL and RF DAC can be used to realize low-phase noise and frequency agile DTRM [1].

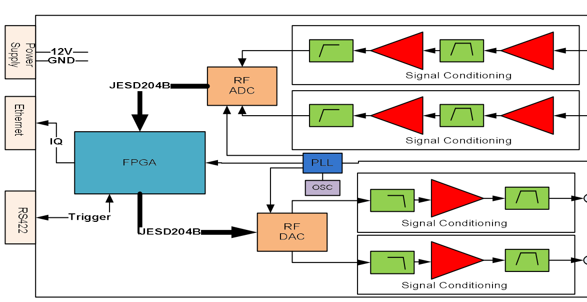

Figure: 1 Block diagram for digital Transceiver

A generic block diagram of the proposed digital transceiver is shown in figure 1. The received signal from the antenna is directly given to the receiving path of DTRM which has a signal conditioning block, RF ADC, and JESD204B link to FPGA. The signal condition block contains a low-noise amplifier (LNA) and a filter. In the transmit path digital baseband signal is given to RF DAC through JESD204B and DAC converts it to RF.

II. CONCEPTUAL DESIGN

Impact of clock noise on an RF-sampling DAC system

The sampling clock of RF DAC plays a very important role in waveform generation. DAC output will generate current noise caused by the sampling clock jitters. The generated output noise will degrade the DAC output's spurious free dynamic range (SFDR) of RF DAC. Zero crossing of a periodic signal in the time domain is the measure of phase noise. Consider a sine wave with phase fluctuations [3],[4]

$$x(t) = \sin(2\pi f t + \varphi(t)) \qquad (1)$$

Where f is frequency, is the random fluctuating phase in radians with case the phase noise. The power spectral density of the phase variations is the measure of phase noise

$$s(f) = \frac{\varphi(t)^2{}_{RMS}}{BW} \quad \frac{rad^2}{Hz} \qquad (2)$$

In linear terms, the phase noise of single sided signal is defined as

$$L(f) = {}^{s(f)}\!/_{2} \qquad (3)$$

Phase noise is expressed in units of dBc/Hz from 10log(L(f)).

Using Eq 1, we can calculate the impact of high-speed clock noise on RF DAC performance:

$$SNR_{SIG}(dB) = SNR_{CLK}(dB) + 20log\left(\frac{\pi}{sin\left(\frac{\pi f_{SIG}}{f_{CLK}}\right)}\right) \quad (4)$$

where f_{SIG} is the signal frequency and f_{CLK} is the frequency of the DAC sampling clock [2]. With the above equation, FCLK plays the important role in DAC phase noise.

III. IMPLENMETATION

A. Phase noise using RF DAC and External PLL

To get ultra-phase noise using RF DAC, in this paper we have discussed two approaches and compared them.

1. Phase noise using DAC internal PLL and low reference clock to DAC as shown in figure 2. In this approach, we need an on-board low clock to DAC and by using DAC internal PLL, DAC sampling frequency is generated. The implementation point of view of this approach is simple but phase noise is less in comparison to our design specifications.

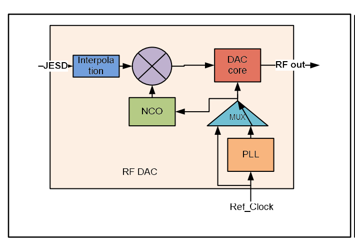

Figure 2: Phase noise calculation using internal PLL for DAC

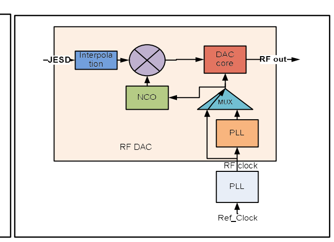

Figure 3: Using external PLL and RF DAC

2. Phase noise using external PLL and DAC is shown in figure3. In this approach, we need to generate a DAC sampling clock using external PLL which is in the order of GHz. From an implementation point of view, this approach is complex but a 10dBc/Hz phase noise improvement is observed as shown in table 1.

B. *Calculation of switching time for agility mode of operation.*

The 48-bit dual modulation mode uses an NCO, a phase shifter, and a complex modulator to modulate the signal by a programmable carrier signal. The frequency tunning word (FTW) for the channel NCOs depends on the speed at which the channel NCO block is running and can be calculated by using the following formulas.

$$FTW = \left(f_{carrier} / f_{DAC}\right) * 2^{48} \qquad (8)$$

FTW registers are not updated immediately upon writing. Instead, the FTW registers updates on the rising edge of DDSC_FTW_LOAD_REQ (Register 0x131, Bit 0). After an update request, DDSC_FTW_LOAD_ACK (Register 0x131, Bit 1) must be high to acknowledge that the FTW has been updated.

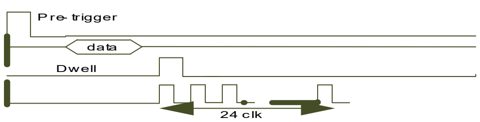

Figure 4: Time control signal

Using the above time control, switching time is measured as 500ns.

IV. RESULTS

The developed DTRM has been tested in the L-band over a 100 MHz bandwidth in 1 MHz frequency steps. The dynamic range of this module is 100dB (input power level from -10 dBm to -110 dBm). Pulse width and PRT can be programmed with firmware to achieve better range resolution. This module has an FPGA and a hard processor for signal processing. Pulse compressed IQ data can be sent over a 1Gb Ethernet link for further processing. 10dBc/Hz phase noise improvement is observed as shown in Table 1 in comparison to the first method and phase noise plots for the above two discussed conditions are captured and shown in figure 6 and figure 7. This compact digital transceiver has replaced the LNA block, waveform generator, timing generator, upconverter, downconverter, and IF digital receiver modules of the existing RADAR and with this innovation, the overall size of the existing RADAR reduced drastically and made this new RADARA a SWaP (Size, weight, and power) system.

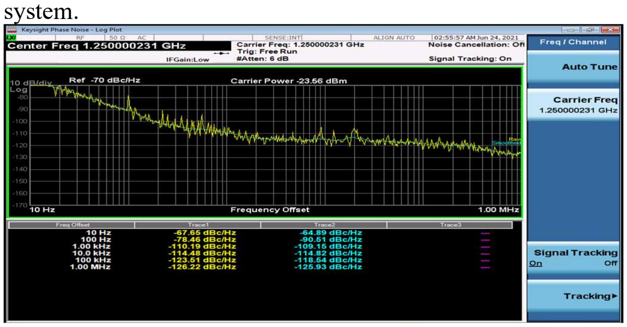

Figure 6: Phase noise plot using internal DAC PLL

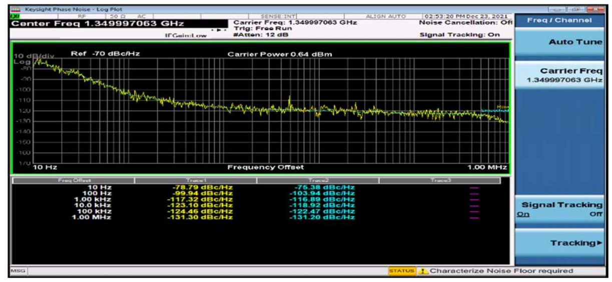

Figure 7: Phase noise plot using external PLL and DAC

Table1: Phase noise comparison

S. N	***Offset frequency***	***Target Spec***	***Using internal PLL***	***Using External PLL***
1	10Hz	-75dBc/Hz	-65dBc/Hz	-78dBc/Hz
2	100Hz	-103dBc/Hz	-90dBc/Hz	-103dBc/Hz
3	1KHz	-116dBc/Hz	-109dBc/Hz	-117dBc/Hz
4	10KHz	-118dBc/Hz	-125dBc/Hz	-119dBc/Hz

In the agility mode, frequency switching time of 500ns has been achieved using direct up-conversion based RF DAC. Received signal is down-converted to the base band using RF ADC and digital pulse compression (DPC) is done inside FPGA.DPC plot is shown in figure 8.

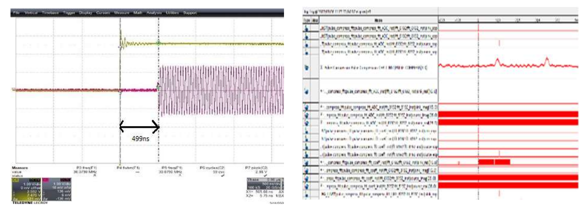

Figure 8: Switching time plot (500ns) and DPC output at sensitivity level (-110 dBm input)

V. CONCLUSION

Digital transceivers have the advantage of improved SNR, configurable amplitude, wide frequency range and low power consumption. Proposed digital transceiver configuration is modular in nature and supports open standard architectures of industry. The dynamic range of this module for the received signal in L-band is 100dB. This paper examines two methods to get ultra-low phase noise for the transmit waveform for the high range resolution pulse radar. Detailed studies and results are shown for digital transceiver fast switching time in the radar agility mode.

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